Bundling is a new overlay protocol to put together a set of heterogeneous Internets. Bundles (messages) are arbitrarily long messages designed for end-to-end delivery between IPN nodes over distinct or identical transport layers. These are routed by a routing function through a concatenated series of Internets, like how terrestrial IP routes packets through a series of independent sub-networks.

To achieve guaranteed end-to-end delivery, bundles may be retransmitted as in Automatic Repeat Request (ARQ) protocol used in TCP layer of terrestrial Internet. The interplanetary backbone of IPN depends on very fragile wireless links. Besides, hubs on the interplanetary backbone move with respect to each other. This may disturb the line-of-sight communication, which needs some sort of relay communication from time to time. Hence, unlike Earth's backbone environment of continuous connectivity, IPN transmission may get discontinued.

Bundles are routed, forwarded and custody-transferred with new concepts. In IPN, routing means using best efforts to send bundles to the next IPN hop to reach the desired destination. Forwarding means sending bundles on demand for non-persistent nodes. Custody transfer is a reliable intra-IPN delivery with storage for persistent nodes. In fact, in custodial transfer, the sender can transmit a file to the receiver over a single link and, on receipt of the entire file, the receiver can notify the sender about any successive forward transmission hops.

IPN nodes are of the following types: agent, relay, gateway and custody transfer.

- Agents build and consume bundles.
- Relays are agents that forward bundles within or between regions.
- Gateways are relays that do routing between regions.
- Custody transfers are optional, and optional versus nodes.

Alternative IPN nodes may be Non-Persistent (NP) node, Persistent (P) node and Exception (E) node. IPN nodes face huge challenges due to instability of backbone and a very long-haul link (long delay and error). An interplanetary backbone is a set of high-capacity, high-availability links between network traffic hubs like terrestrial network backbones. The difference is that hubs and nodes in IPN are, in many cases, hundreds of millions of kilometres apart.

The final journey of the Internet

The Internet is the most amazing gift from the field of information technology (IT). The father of information theory Claude Elwood Shannon defines information from the point of view of communication engineering. In his information theory, more entropy means more information. While in the probabilistic second law of thermodynamics, more entropy means more disorder. Does this mean that more information results in decrease in order? If yes, how?

Entropy of thermodynamics (decrease in order) is a measure of how much a reaction is irreversible. The steam engine produces some waste heat energy. This waste heat energy that causes hot atoms to randomly bounce around is improbable to get back into orderly atoms. Once you get some information, you get so by consuming some energy either by computer processing, network information downloading or other means of communication.

These functions produce some waste heat, which is irreversible. Thus, the durable definition of entropy—a measure of information by Shannon—perfectly matches with entropy of thermodynamics.

An appropriate definition of information in the IT age is by Tom Stonier. Recently, Stonier speculated in his work, titled 'Information and Internal Structure of the Universe', that there is an analogy between mass/matter, energy/heat and information/order of an organisation. He argued that information (I) resident in any organisation is proportional to order (O) of the organisation:

$I = C.O$

where C is the constant of proportionality

If this relation exists, there may be a possibility of interchangeability of information with energy, which, otherwise speaking, will establish a measurable and quantifiable relation between an industrial-based society and an information-based one.

Stonier established an exchange rate, which is one Joule per degree Kelvin=10^{23} bits of information. This has raised a lot of criticism and questions, but there is a direction, which, if proven correct in the future, may lead to the conclusion that information is not external to nature but a fundamental unit of nature.

Shannon's theory demonstrates the inverse relationship between information and entropy in case of communication or information processes. The inverse relationship also holds good for physical or life processes. By the relation between information and knowledge, as demonstrated by several works, it is a reasonably good assumption that knowledge is proportional to information. Hence, entropy and knowledge hold an inverse relationship to each other.

The laws of thermodynamics, particularly the second law, govern